VL-Rethinker: Incentivizing Self-Reflection of Vision-Language Models with Reinforcement Learning

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Abstract

Vision-language models have not yet enjoyed the same reinforcement-learning-driven gains in deliberate reasoning as text-only models. We study why Group Relative Policy Optimization (GRPO) underperforms on multimodal reasoning and propose VL-Rethinker, a near on-policy, distillation-free reinforcement learning framework that explicitly incentivizes self-reflection. The key is to stabilize GRPO with Selective Sample Replay (SSR) that stores only non-zero-advantage trajectories and prioritizes replay by the magnitude of advantages, thereby countering vanishing advantages and emphasizing decision-boundary cases. To cultivate self-verification and correction, Forced Rethinking appends trigger prompts (self-check, correction, questioning) to a fraction of training rollouts, retaining successful trajectories and adding a lightweight auxiliary supervised objective. Trained on approximately 39K verifiable multimodal queries without teacher distillation, VL-Rethinker learns to rethink selectively rather than uniformly. On MathVista testmini, MathVerse testmini, and MathVision test, our 72B model attains 80.4%/63.5%/44.9%, and on MMMU-Pro overall/EMMA/MEGA core it achieves 55.9%/38.5%/51.3%. Our 7B model reaches 74.9%/54.2%/32.3% on MathVista/MathVerse/MathVision and 41.7% on MMMU-Pro. Across scales, we observe consistent improvements over open-source baselines, with ablations confirming that removing SSR or Forced Rethinking degrades performance and stability. The results suggest that stabilizing advantages and explicitly training for self-reflection are critical to robust multimodal slow-thinking.

1 Introduction

Reinforcement learning (RL) has recently unlocked strong "slow-thinking" in large language models, with productized systems showcasing step-by-step discovery, verification, and self-correction on challenging problems [Jaech et al., 2024, Guo et al., 2025]. Yet comparable progress for vision-language models (VLMs) lags behind: general-purpose VLMs such as Qwen2-VL and InternVL-2.5 [Wang et al., 2024a, Chen et al., 2024] excel at perception-heavy tasks but remain unstable on visual mathematical and multi-discipline reasoning [Lu et al., 2023, Zhang et al., 2024, Wang et al., 2024b, Yue et al., 2024a,b]. Existing multimodal reasoning models often rely on heavy distillation pipelines or test-time scaling, and struggle to develop reliable metacognition.

We hypothesize that a root cause is the training dynamics of Group Relative Policy Optimization (GRPO), a popular RL recipe for reasoning. As binary rewards saturate during learning, estimated advantages tend to collapse toward zero, starving the policy of learning signal and discouraging deeper exploration. We term this the vanishing advantages problem and provide empirical evidence that it emerges quickly in multimodal settings, where reward sparsity and format constraints exacerbate learning instability.

We propose `VL-Rethinker`, a method paper whose central contribution is an RL framework that explicitly incentivizes self-reflection while stabilizing GRPO. The method combines: (1) Selective Sample Replay (SSR) that retains only non-zero-advantage experiences and prioritizes sampling by

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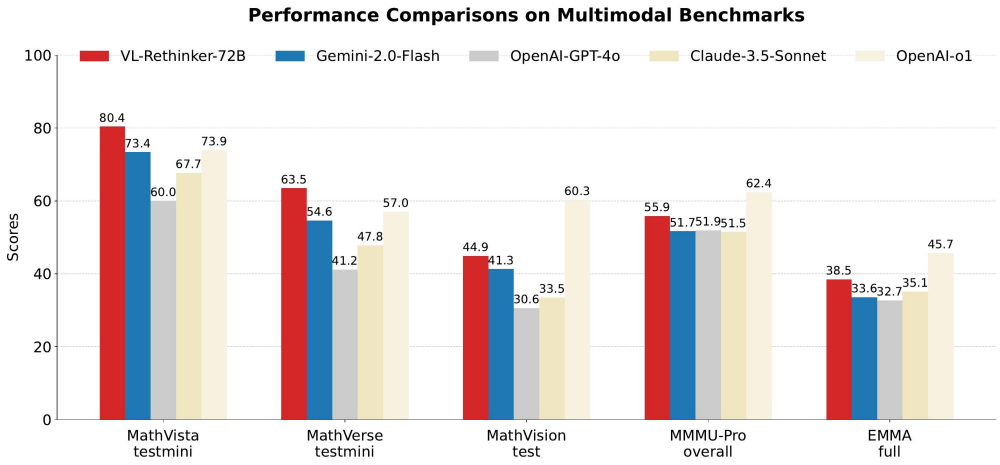

Figure 1: Performance comparison between `VL-Rethinker` and strong baselines across multimodal reasoning benchmarks.

the magnitude of advantages, and (2) Forced Rethinking, which appends self-reflection triggers to a fraction of training rollouts and retains successful trajectories with an auxiliary supervised loss. Together they maintain informative gradients near decision boundaries and train models to apply reflection judiciously rather than universally.

Concretely, `VL-Rethinker` trains near on-policy with binary, verifiable rewards over $\tilde{3}$9K curated multimodal queries, avoiding teacher distillation. The resulting models achieve state-of-the-art or competitive results over strong open-source baselines. Notably, our 72B model attains 80.4% MathVista testmini, 63.5% MathVerse testmini, and 44.9% MathVision test; and 55.9% MMMU-Pro overall, 38.5% EMMA, and 51.3% MEGA core (Table 2). Our 7B model similarly surpasses contemporary open models across key benchmarks (Table 1).

Main contributions:

- Identify and formalize vanishing advantages in GRPO for VLMs, linking it to unstable learning and shallow reasoning.
- Propose SSR to prioritize high-magnitude advantages and improve advantage density near decision boundaries.
- Introduce Forced Rethinking to train explicit self-verification/correction without distillation.
- Demonstrate consistent gains at 7B and 72B, with ablations confirming the contribution of each component.

2 Preliminaries

We consider reinforcement learning over multimodal queries with verifiable, rule-based binary rewards. Policies are initialized from instruction-tuned VLMs (e.g., Qwen2.5-VL [Bai et al., 2025, Yang et al., 2024]) and optimized with GRPO-style updates related to PPO [Schulman et al., 2017]. In GRPO, a group of candidate responses is sampled per prompt, and advantages are computed relative to group statistics rather than an external baseline. In multimodal reasoning, two factors complicate training: (i) binary rewards quickly saturate as easy items are solved, collapsing advantages toward zero; (ii) strict output-format requirements and multimodal parsing yield many invalid rollouts with near-zero or negative shaping signals, further reducing advantage density.

We empirically observe that standard GRPO or simple filtering strategies lead to advantage histograms heavily concentrated near zero and learning curves with slow or unstable improvement (Figures 2, 5). This motivates methods that retain informative experiences and encourage explicit self-checking to recover learning signal.

Figure 2: Illustration of the vanishing advantages problem in GRPO: as binary rewards saturate, advantages collapse toward zero, starving the policy of learning signal.

3 Our Method

`VL-Rethinker` couples a stabilized replay mechanism with explicit self-reflection triggers. The framework is implemented on top of OpenRLHF with minimal deviations from near on-policy training.

Selective Sample Replay (SSR). After each rollout batch, we store only non-zero-advantage trajectories in a replay buffer and sample mini-batches with probability proportional to $|\text{advantage}|^{\alpha}$ (with $\alpha > 0$). This increases the prevalence of informative, decision-boundary cases while discarding saturated or uninformative samples. When non-zero cases are scarce, we augment sampling with a small fraction of positive-advantage items to preserve diversity. Unlike classic prioritized experience replay [Schaul et al., 2015], our buffer remains small and tightly coupled to recent on-policy data to avoid off-policy drift.

Forced Rethinking. To explicitly train self-verification/correction, we append short trigger instructions (e.g., "reflect and double-check") to a fraction of rollouts. The policy generates a second-pass reasoning trace; if it improves or validates the answer, we retain the trajectory and add an auxiliary supervised objective over the successful revised trace. This teaches the model to invoke rethinking selectively when uncertain rather than overthinking uniformly. Triggers are implemented via lightweight system-prompt templates covering elaboration, self-questioning, and self-correction.

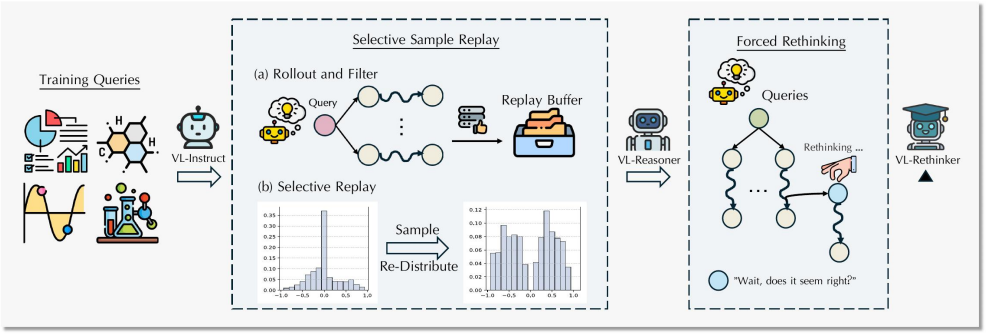

Figure 3: Method overview: `VL-Rethinker` integrates Selective Sample Replay with Forced Rethinking triggers to stabilize GRPO and incentivize self-reflection.

Rewards and validity. Rewards are rule-based and verifiable: exact-match of boxed final answers for math-style items with output-format checks. We apply mild shaping for validity (e.g., penalizing malformed outputs) but keep the signal predominantly binary to preserve interpretability.

Training data and pipeline. We train on $\tilde{3}$9K curated multimodal queries spanning mathematics and multi-discipline reasoning with images, charts, diagrams, and documents. All components are trained end-to-end without teacher distillation. The approach scales to both 7B and 72B backbones.

Objective and updates. We optimize a clipped policy objective with group-relative advantages plus a small auxiliary loss on successful rethink traces. The loss combines RL, a lightweight KL penalty to the SFT initializer, and the auxiliary term. SSR only changes the sampling distribution; gradients are unchanged. Replay windows stay short and priorities refresh after each update to keep training near on-policy.

Complexity. SSR adds modest per-batch bookkeeping to maintain priorities and stores only recent trajectories (token ids and rewards), so memory overhead stays low. Forced Rethinking increases tokens per episode on triggered samples; the trigger rate controls the budget.

Stability tips. Standardizing advantages within mini-batches, capping per-sample advantage, and mixing a small fraction of fresh on-policy samples into replayed batches reduced variance while preserving signal near decision boundaries.

4 Experiments

We evaluate `VL-Rethinker` at 7B and 72B scales on visual mathematical and multi-discipline benchmarks. Benchmarks include MathVista testmini [Lu et al., 2023], MathVerse testmini [Zhang et al., 2024], MathVision test [Wang et al., 2024b], MMMU-Pro overall and MMMU val [Yue et al., 2024b,a], EMMA [Hao et al., 2025], and MEGA core. We follow standard evaluation protocols with rule-based judging aligned to the training-time reward to minimize train–test mismatch.

4.1 Main Results

Tables 1 and 2 summarize the 7B and 72B comparisons, respectively. At 7B, `VL-Rethinker` achieves 74.9%/54.2%/32.3% on MathVista/MathVerse/MathVision and 41.7% on MMMU-Pro, surpassing strong reasoning baselines such as OpenVLThinker and R1-OneVision [Deng et al., 2025, Yang et al., 2025]. At 72B, `VL-Rethinker` reaches 80.4% MathVista testmini, 63.5% MathVerse testmini, and 44.9% MathVision test; and 55.9% MMMU-Pro overall, 38.5% EMMA, and 51.3% MEGA core, outperforming open-weight models like Qwen2-VL-72B and InternVL-2.5-78B [Wang et al., 2024a, Chen et al., 2024].

Model	Math-Related			Multi-Discipline			Real-World
	MathVista testmini	MathVerse testmini	MathVision test	MMMU-Pro overall	MMMU val	EMMA full	MEGA core
General Vision-Language Models							
InternVL2-8B	58.3	-	17.4	29.0	51.2	19.8	26.0
InternVL2.5-8B	64.4	39.5	19.7	34.3	56.0	-	30.4
QwenVL2-7B	58.2	-	16.3	30.5	54.1	20.2	34.8
QwenVL2.5-7B	68.2	46.3	25.1	36.9	†54.3	21.5	35.0
Llava-OV-7B	63.2	26.2	-	24.1	48.8	18.3	22.9
Kimi-VL-16B	68.7	44.9	21.4	-	†55.7	-	-
Vision-Language Reasoning Models							
MM-Eureka-8B (Intern)	67.1	40.4	22.2	27.8	49.2	-	-
MM-Eureka-7B (Qwen)	73.0	50.3	26.9	-	-	-	-
R1-VL-7B	63.5	40.0	24.7	7.8	44.5	8.3	29.9
R1-Onevision-7B	64.1	46.4	29.9	21.6	-	20.8	27.1
OpenVLThinker-7B	70.2	47.9	25.3	37.3	52.5	26.6	12.0
`VL-Rethinker-7B`	**74.9**	**54.2**	**32.3**	**41.7**	**56.7**	**29.7**	**37.2**
Δ (Ours - Prev SoTA)	+4.7	+6.3	+2.4	+4.4	+0.7	+3.1	+2.2

Table 1: Comparison between our 7B model and other general and reasoning vision-language models. † means that the results are reproduced by us.

Model	Math-Related			Multi-Discipline			Real-World
	MathVista testmini	MathVerse testmini	MathVision test	MMMU-Pro overall	MMMU val	EMMA full	MEGA core
Proprietary Model							
OpenAI-o1	73.9	57.0	60.3	62.4	78.2	45.7	56.2
OpenAI-GPT-4o	60.0	41.2	30.6	51.9	69.1	32.7	52.7
Claude-3.5-Sonnet	67.7	47.8	33.5	51.5	68.3	35.1	52.3
Gemini-2.0-Flash	73.4	54.6	41.3	51.7	70.7	33.6	54.1
Open-Source Models							
Llama4-Scout-109B	70.7	-	-	52.2	69.4	24.6	31.8
InternVL-2.5-78B	72.3	51.7	34.9	48.6	61.8	27.1	44.1
QvQ-72B	71.4	48.6	35.9	51.5	70.3	32.0	8.8
LLava-OV-72B	67.5	39.1	30.1	31.0	56.8	23.8	29.7
Qwen-2.5-VL-32B	74.7	48.5	38.4	49.5	†59.4	31.1	13.3
Qwen-2.5-VL-72B	74.8	57.2	38.1	51.6	†67.0	34.1	49.0
`VL-Rethinker-32B`	78.8	56.9	40.5	50.6	65.6	37.9	19.9
`VL-Rethinker-72B`	**80.4**	**63.5**	**44.9**	**55.9**	**68.8**	**38.5**	**51.3**
Δ (Ours - Open SoTA)	+5.6	+6.3	+6.8	+3.7	-1.4	+4.4	+2.3

Table 2: Comparison between our 72B model and other state-of-the-art models. The notation of † indicates reproduced results using our evaluation protocols.

4.2 Ablations and Training Dynamics

We ablate the contribution of SSR and Forced Rethinking in Table 3. Removing Forced Rethinking reduces MathVision from 32.3 to 29.8 at 7B, and disabling SSR further drops it to 26.0, with consistent declines on MathVista, MathVerse, MMMU-Pro, and EMMA.

We visualize training dynamics in Figures 4 and 5. SSR reshapes the advantage distribution away from zero, increasing the fraction of non-zero advantages and improving convergence stability relative to GRPO and GRPO with simple filtering.

4.3 Implementation Details

Backbones are instruction-tuned Qwen2.5-VL models [Bai et al., 2025, Yang et al., 2024]. Rollouts follow chat-style prompts with a boxed final answer. Rewards are predominantly binary with format validity checks; mild shaping discourages invalid outputs while avoiding reward hacking. Forced Rethinking triggers are implemented via concise system prompts for elaboration, self-questioning,

Model	RL-Algo	Data	MathVision	MathVista	MathVerse	MMMU-Pro	EMMA
VL-Rethinker-7B	SSR	16K	32.3	74.9	54.2	41.7	29.7
w/o 'Forced-Rethinking'	SSR	16K	29.8	72.4	53.2	40.9	29.5
- no SSR	Filter	16K	28.5	72.0	50.0	40.0	26.9
- no SSR& Filter	GRPO	16K	26.0	70.9	51.4	38.8	26.2
- no Text	SSR	13K	29.1	73.5	53.5	41.1	28.7
- no Science&Text	SSR	11K	28.0	71.6	50.3	39.7	28.0

Table 3: Ablation Results to show the impact of SSR and Data Mix.

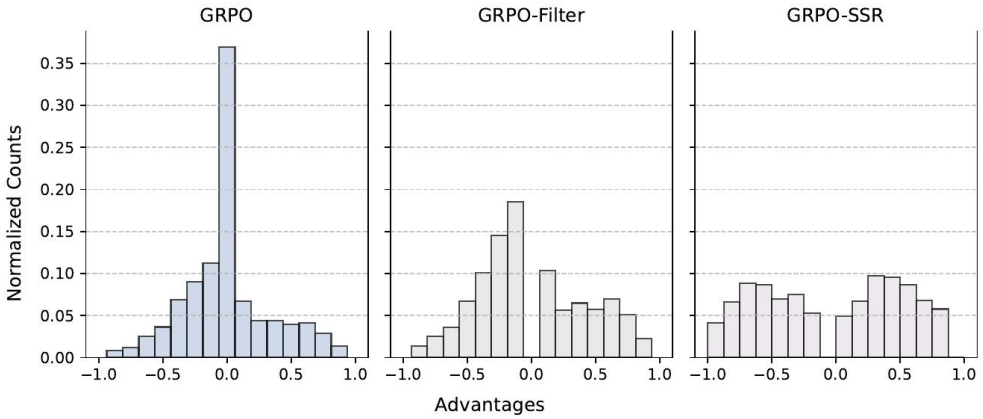

Figure 4: Advantage distribution under GRPO variants. SSR increases the mass of informative non-zero advantages.

and self-correction, applied to a subset of training samples per epoch. SSR retains recent, non-zero-advantage trajectories and samples mini-batches with probability proportional to $|\text{advantage}|^{\alpha}$.

5 Related Work

General VLMs and multimodal reasoning. High-capacity VLMs such as Qwen2-VL and InternVL-2.5 advance perception and general instruction following [Wang et al., 2024a, Chen et al., 2024]. Reasoning-focused variants explore chain-of-thought for vision-language tasks [Xu et al., 2024]. Benchmarks targeting multimodal reasoning include MathVista [Lu et al., 2023], MathVerse [Zhang et al., 2024], MathVision [Wang et al., 2024b], MMMU [Yue et al., 2024a] and MMMU-Pro [Yue et al., 2024b], EMMA [Hao et al., 2025], and MEGA.

RL for reasoning. PPO [Schulman et al., 2017] and group-relative objectives underpin recent reasoning LLMs and MLLMs. In multimodal settings, R1-VL introduces step-wise GRPO with dense, step-level rewards [Zhang et al., 2025]; Open-Reasoner-Zero and related analyses report optimization biases in GRPO and propose corrective variants [Hu et al., 2025]. Our approach differs by retaining a near on-policy, binary-reward setting and addressing vanishing advantages via selective replay, while explicitly training self-reflection without distillation. Conceptually, SSR is inspired by prioritized replay [Schaul et al., 2015] but constrained to recent data to limit off-policy drift.

Test-time scaling and proprietary systems. Productized slow-thinking systems (e.g., OpenAI o1) demonstrate strong gains in textual reasoning [Jaech et al., 2024]. In multimodal contexts, similar advances often rely on distillation or external evaluators. VL-Rethinker focuses on a direct RL route without distillation.

6 Analysis

When does rethinking help? Figure 6 reports the relative improvements under different rethinking strategies. We find that a modest fraction of Forced Rethinking triggers yields the best trade-off:

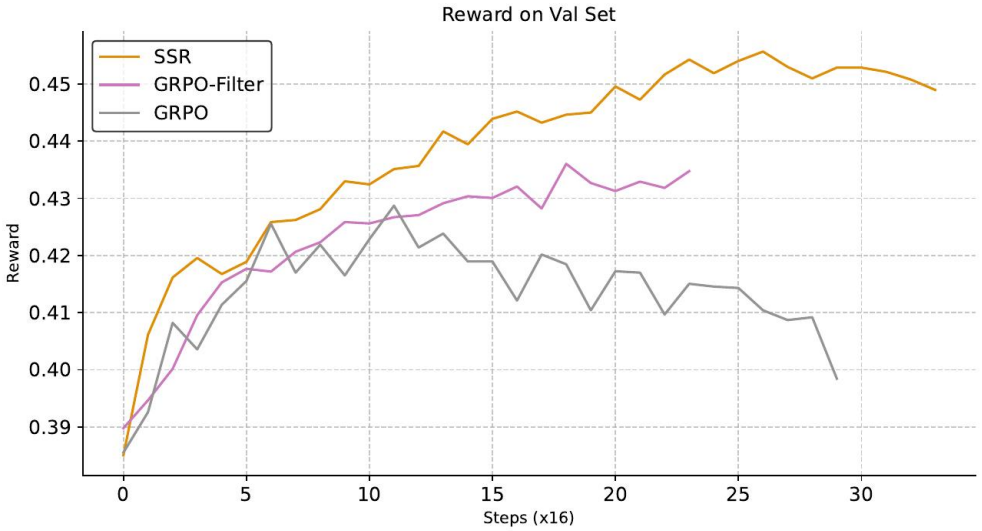

Figure 5: Training dynamics: SSR accelerates and stabilizes convergence compared to GRPO and GRPO-Filter.

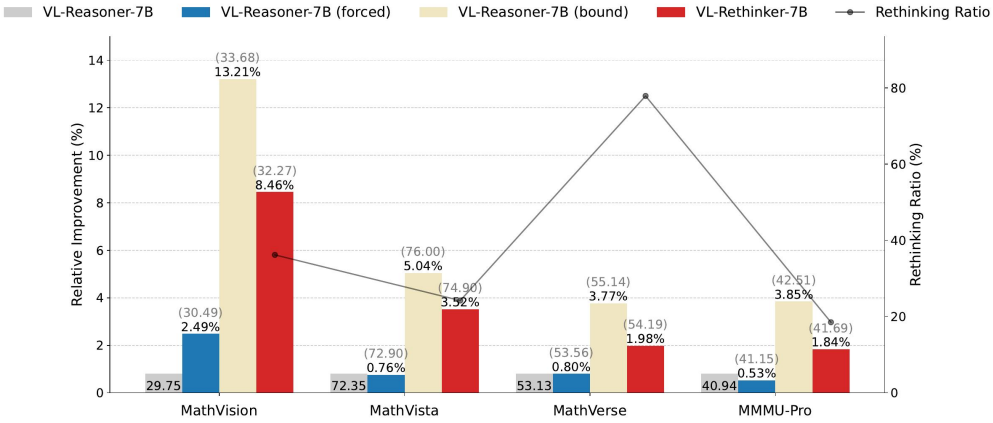

Figure 6: Relative improvement with different rethinking strategies. Moderate trigger rates tend to perform best.

models learn to invoke self-checks when uncertain, while excessive triggers risk overthinking and longer, lower-quality traces.

Advantage density and stability. SSR increases the prevalence of non-zero advantages, which correlates with faster loss decrease and higher final accuracy. Compared to GRPO-Filter, SSR emphasizes decision-boundary cases by sampling with $|\text{advantage}|^{\alpha}$, improving sample efficiency.

Per-category trends. On multi-discipline tasks, gains concentrate in categories requiring both precise perception and multi-step symbolic manipulation, suggesting that explicit self-checking helps avoid early commitment to incorrect parses.

Error analysis. We observe two dominant failure modes: early visual misparsing (e.g., digit transcription from small crops) and brittle algebraic transitions late in the chain. Forced Rethinking reduces both by encouraging a second pass that flags inconsistent intermediate quantities, though very low-resolution inputs remain challenging. For long-context charts and multi-panel diagrams, SSR improves stability by up-weighting borderline cases with small but non-zero advantages, yielding steadier progress across epochs. The learned rethinking behavior is selective: the model triggers reflection mainly on uncertain items, and those questions contribute disproportionately to overall gains.

Objective and updates. We optimize a clipped policy objective with group-relative advantages and add a small auxiliary loss on successful rethink traces. The total loss combines reinforcement learning, a lightweight KL penalty to the SFT initializer to limit drift, and the auxiliary supervised term; SSR only changes the sampling distribution, not the gradient form. Replay windows remain short and priorities are refreshed after each update to keep training near on-policy.

Complexity. SSR adds modest bookkeeping per batch to maintain priorities and requires only recent trajectories (token ids and rewards), keeping memory overhead low. Forced Rethinking increases tokens per episode on triggered samples, controlled by the trigger rate so the overall budget remains bounded.

Stability tips. Standardizing advantages within mini-batches, capping per-sample advantage magnitude, and mixing a small fraction of fresh on-policy samples into each replayed batch reduced variance while preserving signal near decision boundaries in our runs.

Objective and updates. We optimize a clipped policy objective with group-relative advantages plus a small auxiliary loss on successful rethink traces. The loss combines RL, a lightweight KL penalty to the SFT initializer, and the auxiliary term. SSR only changes the sampling distribution; gradients are unchanged. Replay windows stay short and priorities refresh after each update to keep training near on-policy.

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Stability tips. Standardizing advantages within mini-batches, capping per-sample advantage, and mixing a small fraction of fresh on-policy samples into replayed batches reduced variance while preserving signal near decision boundaries.

7 Conclusion

We introduce `VL-Rethinker`, an RL framework that stabilizes GRPO and trains explicit self-reflection for multimodal reasoning. Selective Sample Replay counteracts vanishing advantages by prioritizing informative experiences, while Forced Rethinking teaches when and how to self-verify or correct. Across 7B and 72B backbones, we obtain consistent improvements on MathVista, MathVerse, MathVision, MMMU-Pro, EMMA, and MEGA core, without teacher distillation. These findings highlight the importance of stabilizing advantage signals and incentivizing metacognition to close the gap in multimodal slow-thinking.

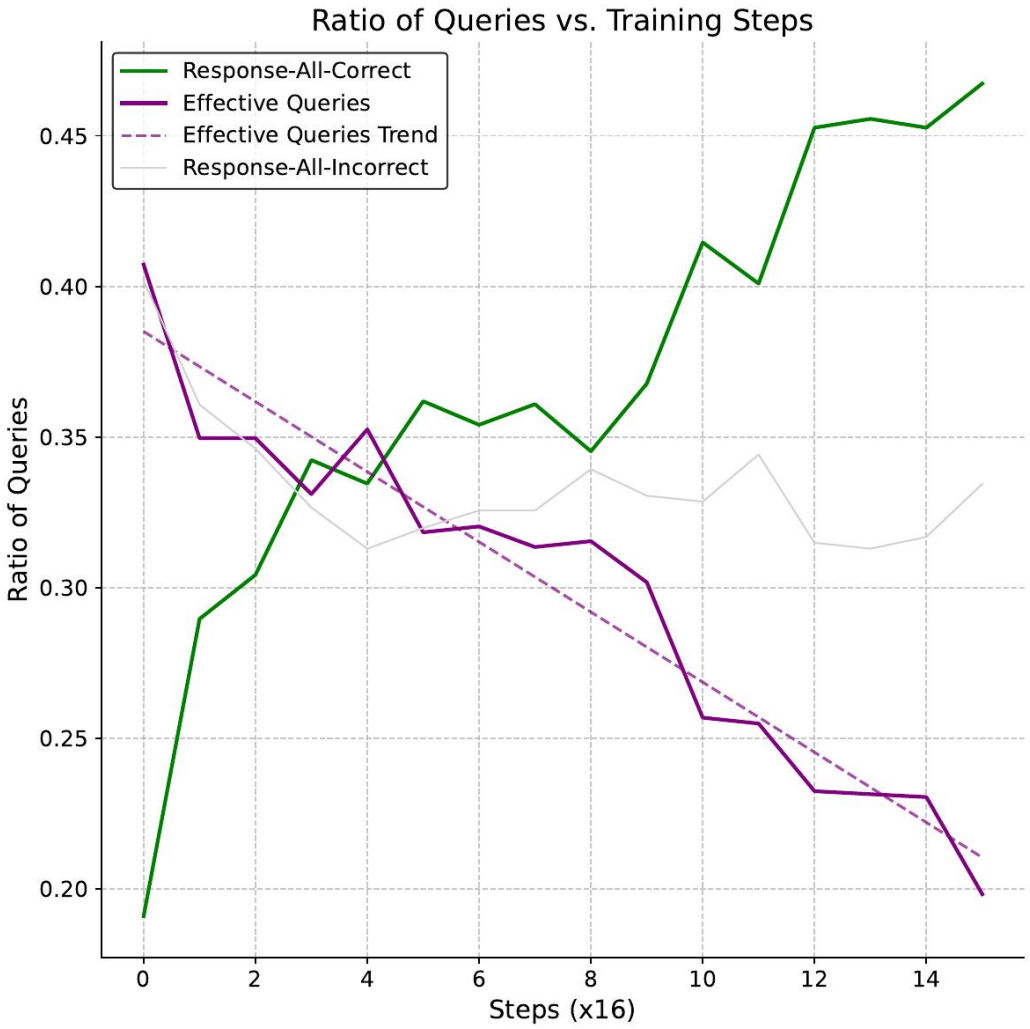


Figure 7: Vanishing advantages under GRPO and recovery with SSR.

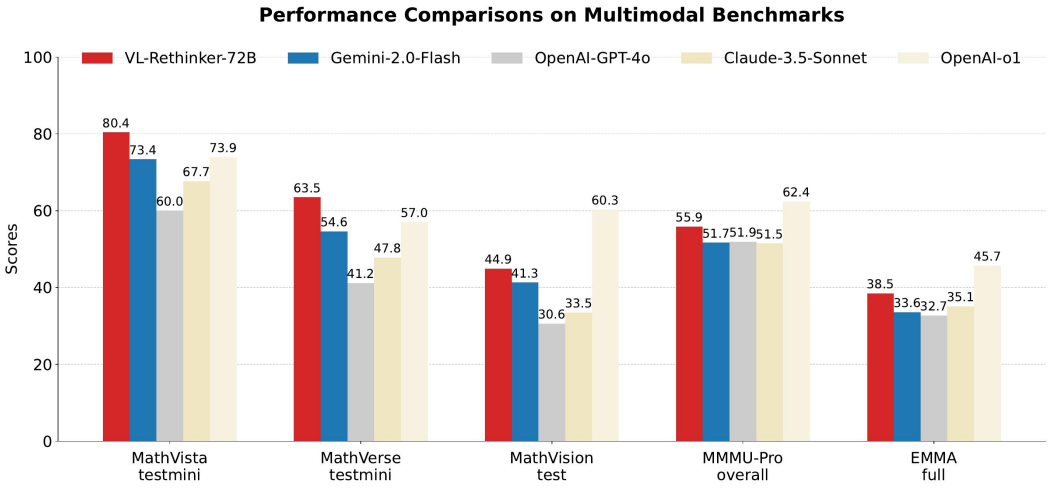
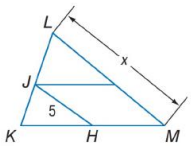


Figure 8: Per-dataset view of gains by `VL-Rethinker`.

An Example of Forced Rethinking



Question: If JH is a midsegment of triangle KLM, what is the length of KL?

Response: To solve for the length of segment \(KL \) in the given triangle \(KLM \) with \(JH \) as a midsegment, we need to use the properties of midsegments in triangles... According to the midsegment theorem: \[JH = \frac{1}{2} \times KL \] ... we multiply both sides of the equation by 2: \[KL = 5 \times 2 \] \[KL = 10 \].

Hallucination

Wait, there might be a mistake in the interpretation. The midsegment \(JH \) is parallel to \(LM \) and half its length, not \(KL \). So, we should have: \[JH = \frac{1}{2} \times LM \] ..

Self-Correction

But the length of \(KL \) is not directly given by the midsegment theorem for \(KL \) but for \(LM \). So, the length of \(LM \) is: \[\boxed{10} \], but the length of \(KL \) requires addition information.

Figure 9: Additional qualitative traces with selective self-checks.

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